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COMPUTER
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SMART-C:
Smart Management and Reservation Tool for Campus

Bachelor's Thesis

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ABSTRACT

SMART-C (Smart Management and Reservation Tool for Campus) refers to an intelligent tool that has been developed to facilitate better management of transportation and facilities reservations on university campuses. Students in several university campuses currently lack information about the occupancy and schedule of shuttles and availability of facilities. Current systems are fragmented and do not offer a unified system for managing such services. This might cause students to wait longer than necessary for these services.

SMART-C stands for Smart Management and Reservation Tool for Campus. SMART-C is an innovative smart campus management system designed for the purpose of making campus management easy and smooth by providing proper information about transportation systems and facility reservations. Many universities do not have any such systems through which they can keep track of the occupancy level of shuttle buses and the availability status of different facilities. This leads to unnecessary delays and wastage of resources.

From the test results, it can be seen that the system was successful in providing features for authentication, reservation management, shuttles' details, and message delivery through a single platform. Conflicts in reservations will not arise due to validation in the back end, while access control makes sure of the safety. It can be observed that the application of a single smart campus platform increases efficiency and accessibility on campus.

ACKNOWLEDGEMENTS

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1. INTRODUCTION

1.1. Problem Definition

The university campus is populated by thousands of students that require different types of facilities including transportation facilities, studying venues, sporting venues, laboratory space, and places for conducting events. Unfortunately, most of these facilities operate within fragmented modes of communications and manual ways of information management, thereby causing problems to students as regards access to the latest information concerning the aforementioned facilities.

Transportation planning is one of the biggest problems faced by universities. The problem that exists here is that students normally do not have any idea about the present capacity of campus buses. Delays, rescheduling of buses, or even overcrowding can cause problems like loss of time. Another aspect in this regard is that no reservation system is normally there regarding campus facilities and this can create conflict for students.

With recent advancements in mobile technology and smart campus applications, it has become possible to solve these problems using a unified digital platform. This involves the integration of transportation management, reservation of facilities, and notifications within one application to enhance efficiency and provide an improved user interface for the students.

In an effort to mitigate the above challenges, a new system referred to as SMART-C (Smart Management and Reservation Tool for Campus) has been formulated. The SMART-C application allows students to access their campus shuttle information, facilities booking and other related campus activities via a mobile application that combines all these features into one platform.

1.2. Project Objectives, Scope, and Contributions

1.2.1. Project Objectives

One of the main goals of SMART-C is to develop a digital interface that can help to facilitate campus transport and facility services. Students are normally constrained to use different means to access information about transport and facility services and receive important announcements. The solution proposed by SMART-C would combine different campus services into one mobile application.

The application will be developed in such a way that students could track occupancy levels of campus transport, reserve campus facilities, manage reservations, and receive important notifications about campus facilities. Thus, the goal of the SMART-C team is to increase accessibility and minimize uncertainties for the benefit of university students.

Another goal of this project is to make students more informed about campus transport by helping to estimate its occupancy level.

1.2.2. Project Scope

The scope of SMART-C includes the development of a mobile-based smart campus management system that supports transportation monitoring, facility reservation management, notification services, and administrative operations within a university campus environment.

The system consists of a Flutter-based mobile application, a Spring Boot backend, and a relational database management system. Through the mobile application, students can access shuttle occupancy information, reserve campus facilities, manage their reservations, receive notifications, and view campus announcements. Administrators can manage shuttle information, facility data, announcements, and exam reservations through authorized system functions.

The transportation module focuses on estimating shuttle occupancy levels using location information obtained from authenticated users located near predefined shuttle departure points. The facility management module enables students to reserve available campus facilities while preventing reservation conflicts through backend validation mechanisms.

In addition, the system includes an administrator-only exam reservation module that supports classroom allocation and prevents scheduling conflicts between examinations. Notification and announcement services provide timely communication between university administrators and students.

The scope of this project is limited to shuttle occupancy estimation, facility reservation management, exam scheduling support, notification services, and related administrative operations. Features such as direct GPS tracking of shuttles, payment processing, and integration with external university information systems are outside the scope of the current implementation and may be considered in future versions.

1.2.3. Location-Based Shuttle Occupancy Estimation

One of the main advantages provided by SMART-C technology is the shuttle occupancy estimation based on location. Unlike the traditional way of estimating the shuttle occupancy level by observing or counting passengers directly, SMART-C technology estimates shuttle occupancy levels through location information of authorized users.

To accomplish this, SMART-C compares the total number of users inside pre-defined shuttle departure locations and estimates their occupancy level comparing it with the capacity of the particular shuttle. This estimate is then displayed to students via the mobile application.

Such an opportunity will allow students to know the crowd levels on a specific shuttle even before arriving at the departure location and making more informed decisions. Additionally, this method allows us to suggest a solution that does not require any additional hardware.

1.2.4. Smart Reservation Management System

Furthermore, SMART-C consists of a reservation management component, which facilitates the reservation of campus facilities, including study rooms, laboratories, sports zones, and other venues using the mobile app.

This reservation mechanism ensures availability verification and conflict-free reservation of facility slots. In addition, students can easily view the list of facilities, select available times, make reservations, and modify previous reservation arrangements.

Moreover, apart from general facility reservation, SMART-C has another capability, which is the reservation of exam rooms. This activity can be conducted exclusively by authorized users and entails setting up examinations for certain classrooms. In fact, one can avoid scheduling conflicts for exam reservations in case of simultaneous reservation of several exam rooms in the same room. This function helps in avoiding scheduling issues, making classroom management more systematic and efficient.

1.2.5. Notification and Announcement Management

A combination of notifications and announcements feature is built into SMART-C to make sure that users get the most current and relevant information on the availability of campus facilities.

Notifications related to key events, like reservation confirmation, reservation cancellation, shuttle schedule and other announcements, can be provided by SMART-C. They enable users to get the latest information from the campus.

Also, the announcement capability of SMART-C makes the communication process more effective as far as the administration and users are concerned.

1.2.6. Technical and Architectural Contributions

From a technical point of view, SMART-C uses a contemporary client-server architecture that consists of a mobile application built with Flutter technology, a Spring Boot backend, and a database management system.

The mobile app is the user interface in this case, while the backend will be used to authenticate users, make estimates of the occupancy of the shuttle bus, manage reservations and notifications, and perform data processing. Communication between different components of the system is provided by means of RESTful APIs.

A modular architecture used in SMART-C makes this system maintainable, scalable, and more flexible in terms of possible future developments.

1.2.7. Contribution to Sustainable Development Goals

The SMART-C platform structurally aligns with the United Nations Sustainable Development Goals (SDGs) by transforming physical campus resources and transit services into an integrated, efficient, and data-driven ecosystem. Through its core functionalities, the project directly addresses three primary global goals:

- Goal 11: Sustainable Cities and Communities: By optimizing campus shuttle management and minimizing passenger waiting times, SMART-C mitigates overcrowding and unnecessary transit delays. This contribution promotes sustainable micro-transportation models within the university ecosystem and assists in reducing the overall institutional carbon footprint.
- Goal 12: Responsible Consumption and Production: The system establishes a conflict free, centralized framework for reserving university assets such as study rooms, laboratories, and sports zones. By managing availability dynamically, SMART-C directly prevents the underutilization of campus spaces and resolves operational inefficiencies, successfully addressing the critical issue of resource wastage.

- Goal 9: Industry, Innovation, and Infrastructure: SMART-C replaces traditional, manual, and fragmented information management practices with an advanced digital infrastructure. Utilizing a contemporary client-server architecture comprising a Flutter based mobile interface, a Spring Boot backend, and a relational database the project fosters innovation by providing an integrated smart campus solution.

1.3. Organization of the Thesis

The structure of this thesis consists of six chapters.

In Chapter 1, the problem addressed by this thesis is introduced, followed by the discussion of the objectives, scope, and contribution of SMART-C.

In Chapter 2, the relevant literature is reviewed, focusing on various smart campus applications. Various existing technologies and systems pertaining to campus transportation management, facility reservation system, and smart campuses are examined and compared.

Chapter 3 covers the proposed SMART-C system. Detailed descriptions of the proposed system in terms of the system architecture, graphical user interface, business logic, and data management will be included in this chapter.

Chapter 4 discusses the implementation of the system. Specific technologies, frameworks, and tools used in developing the system will be covered in this chapter.

Chapter 5 describes the test and evaluation of the system. Functionality and non-functionality tests are discussed to prove that the system meets the design requirements.

In conclusion, Chapter 6 includes the summary and discussion on the results of this project, its limitations, as well as future suggestions.

2. LITERATURE REVIEW

This chapter reviews existing systems and technologies related to campus transportation management, facility reservation systems, and smart campus applications. The purpose of this review is to examine current solutions that address problems similar to those targeted by SMART-C and to identify the features that can contribute to a more efficient campus management experience. The reviewed applications focus on transportation information services, reservation management, and centralized access to campus-related services.

2.1. Similar Applications

2.1.1. MazeMap

MazeMap is an online campus navigation management tool which has gained widespread popularity among universities and educational organizations. MazeMap offers interactive campus maps, indoor navigation, classroom searches, and detailed information regarding campus buildings. Students and visitors can navigate the entire campus by searching for rooms and buildings using a mobile or web application.

Advantages of Using MazeMap Navigation System

The key advantage of this system is its ability to centralize campus-related information and make it accessible to all interested parties. With the help of real-time navigation services, users have an easy time moving from one part of the campus to another.

Limitations of Using MazeMap

However, one of the key limitations of MazeMap is its lack of comprehensive facility reservation and shuttle occupancy monitoring services. Thus, while this system greatly enhances the accessibility of campus areas, it fails to meet transportation management needs].

2.1.2. Appointy Scheduling

Appointment Scheduling refers to an internet based scheduling program that can be used in various establishments such as educational institutions, organizations, health care facilities, and even services companies. Users are able to see which time slots are available, book a slot, cancel appointments, and get notifications about their appointments.

This system incorporates the following functionalities: calendar management, scheduling automation, and reservation conflict resolution.

Despite its strong reservation capabilities, Appointy is designed as a general-purpose scheduling solution rather than a campus-specific management platform. It does not provide

transportation-related services, shuttle information, or campus-oriented administrative functions. As a result, it addresses only a subset of the requirements targeted by SMART-C .

2.1.3. Skedda

The Skedda app is used for reserving space, be it work or sports facilities. The platform provides options for booking a room for meetings, for studying, for playing various sports, and any other shared space.

In terms of features, there are possibilities of availability control, approval of bookings, conflict prevention, and reservation tracking. Facility usage policy settings and reservation management capabilities are provided for the administrator side of the system.

Although Skedda provides advanced reservation management capabilities, it focuses exclusively on facility booking operations. Transportation services, shuttle occupancy information, and campus-wide communication features are not included within its scope. Therefore, while Skedda addresses facility management challenges effectively, it does not provide an integrated campus management solution.

2.1.4. Comparison of Similar Applications

Tablo 1. presents a comparison of the reviewed applications and SMART-C based on their major functionalities.

Tablo 1. Comparison of Similar Applications

Features	MazeMap	Appointy Schedulin	Skedda	SMART-C
Mobile Application	✓	✓	✓	✓
Campus-Oriented Design	✓	✗	✗	✓
Shuttle Information Management	✗	✗	✗	✓
Shuttle Occupancy Monitoring	✗	✗	✗	✓
Facility Reservation	✗	✓	✓	✓
Reservation Conflict Prevention	✗	✓	✓	✓
User Authentication	✓	✓	✓	✓
Notifications and Announcements	Limited	✓	✓	✓
Administrative Management Functions	Limited	✓	✓	✓
Integrated Campus Services	✗	✗	✗	✓

As shown in Tablo 1, existing systems generally focus on a single service area. MazeMap emphasizes navigation and campus accessibility, while Appointy and Skedda mainly focus on reservation management. In contrast, SMART-C integrates transportation services, facility

reservations, announcements, notifications, and administrative operations within a single platform.

2.2. Smart Campus Systems

Smart campus systems represent the digital transformation of university environments through the integration of information and communication technologies. These systems aim to improve operational efficiency, resource utilization, accessibility, and communication between students, staff, and administrators].

Recent studies indicate that smart campus platforms can significantly improve campus services by centralizing information and automating routine processes. Services such as transportation management, facility reservations, access control, event management, and notifications are commonly integrated into smart campus solution.

A key characteristic of smart campus systems is the ability to provide real-time information. Students can access up-to-date information regarding campus services through mobile devices, enabling faster decision-making and improved user experience.

Furthermore, centralized systems reduce administrative workload by automating repetitive tasks and improving resource allocation.

Research indicates that centralized campus platforms improve operational efficiency, information accessibility, and resource utilization while reducing administrative workload..

The SMART-C project follows the smart campus concept by integrating multiple campus services into a single mobile application. Transportation information, facility reservations, notifications, announcements, and administrative functions are managed through one centralized platform, improving accessibility and efficiency for all users.

2.3. Campus Transportation Systems

Transportation services play a crucial role in university campuses. Students rely on shuttle services to travel between campus locations and surrounding areas. However, traditional transportation management approaches often provide limited information regarding shuttle availability, occupancy levels, delays, or schedule changes.

Modern transportation systems attempt to solve these problems by providing real-time route information, vehicle tracking, occupancy estimation, and service notifications.. Such systems allow users to make informed decisions before arriving at transportation points, reducing waiting times and improving service utilization.

Several transportation management studies emphasize the importance of timely information delivery.. Students who receive accurate transportation updates are better able to organize their schedules and avoid unnecessary delays. Real-time occupancy information is particularly useful because it helps users estimate crowd levels before using transportation services.

Several studies emphasize that real-time transportation information significantly improves user satisfaction and enables more effective transportation planning within university campuses.

SMART-C addresses these transportation challenges by providing shuttle-related information through a mobile platform. Users can access route information, occupancy data, schedule updates, and transportation announcements from a single interface. This centralized approach improves transportation planning and contributes to a better campus experience.

2.4. Facility Reservation Systems

Facility reservation systems are designed to manage shared resources such as study rooms, laboratories, sports facilities, classrooms, and event halls. These systems help institutions maximize resource utilization while preventing reservation conflicts.

One of the most important requirements of reservation systems is availability management. Users must be able to view available resources and reserve appropriate time slots. In addition, reservation systems must prevent duplicate bookings and ensure data consistency through validation mechanisms].

Modern reservation systems often include features such as booking management, cancellation options, availability monitoring, confirmation notifications, and administrative controls. These capabilities improve resource allocation and reduce scheduling errors].

Reservation validation and conflict prevention mechanisms are considered critical components of modern facility management systems because they improve resource utilization and prevent scheduling conflicts..

The reservation module of SMART-C follows these principles by allowing users to browse facilities, select available time slots, create reservations, and manage existing bookings. Backend validation mechanisms ensure that conflicting reservations cannot be created. This contributes to efficient facility usage and improves the overall user experience.

2.5. Summary of Literature and Research Gap

The above-mentioned applications and studies show that transportation management, facility reservation management, and campus information systems are important aspects of the modern university ecosystem. Nonetheless, most of the applications available today can be used for a particular aspect of the mentioned areas.

MazeMap can be considered a navigation service, whereas Appointy and Skedda provide facility reservation management. Though the mentioned applications are successfully solving their particular areas, they are still unable to combine all of the listed aspects into a single platform.

Besides, most of the existing studies are dedicated either to the management of transportation services or facilities separately. There is a lack of scientific work on how to integrate such services in a single mobile application in order to manage the activities of students. Thus, SMART-C is able to contribute to the science in this way.

Thus, there is a gap in the research on the subject of the development of a comprehensive smart campus system. This research problem is going to be addressed through SMART-C which will offer a consolidated platform where campus services can be managed through a single mobile app backed up by centralized architecture. Through the inclusion of the transportation services, facility reservation, notifications, announcements, and administrative functions, SMART-C will be able to fill in the gap mentioned above.

To conclude, the review of the existing literature proves the increased need for digital campus management systems. The existing systems have been doing a great job at tackling various aspects of campus management. At the same time, it is clear that there is a demand for a consolidated platform offering a wide range of campus management options.

Based on the limitations identified in existing smart campus, transportation, and reservation systems, SMART-C has been developed as an integrated platform that combines these services within a single mobile application. The following chapter presents the architecture, components, and design of the proposed SMART-C system.

3. PROPOSED SYSTEM

This chapter presents the proposed SMART-C system, including its overall architecture, graphical user interface, business logic, and data management structure. SMART-C is designed as a smart campus management platform that integrates shuttle information, facility reservations, occupancy monitoring, announcements, and notification services into a single application.

3.1. Introduction

SMART-C (Smart Management and Reservation Tool for Campus) is designed to supply a combined platform for managing and organizing campus transformation and facility reservation services. Students find it challenging to effectively obtain campus announcements, facility availability, and transit information because current campus services are typically dispersed across several communication channels.

To address these issues, SMART-C integrates transportation management, facility reservation, announcement delivery, and notification services into a single mobile application.A MySQL relational database, a Spring Boot backend, and a Flutter-based mobile application make up the system's client-server architecture.

The mobile application can be used by students , administrators (admins and shuttle admins). The backend handles authorization, authentication , validation of reservations and managements of shuttles and classes , and the notification processings. The mobile application and the backend communicate using RESTful APIs that are protected by JWT authentication.

The SMART-C system follows the MVC (Model-View-Controller) architectural pattern for being modular , maintainable and scalable. In this MVC architecture , Flutter app has a role as View Layer to provide user interaction and showing information. As the Controller layer, the system has Spring Boot REST controllers that takes requests from application and organizes ,coordinates operations. The Model layer consists of entity classes and database structures which show users, reservations, facilities, shuttles, announcements and notifications. Business rules acts as an intermediary between controllers and database.

This separation of concerns boots code organization, makes easier maintenance and supports future systems.

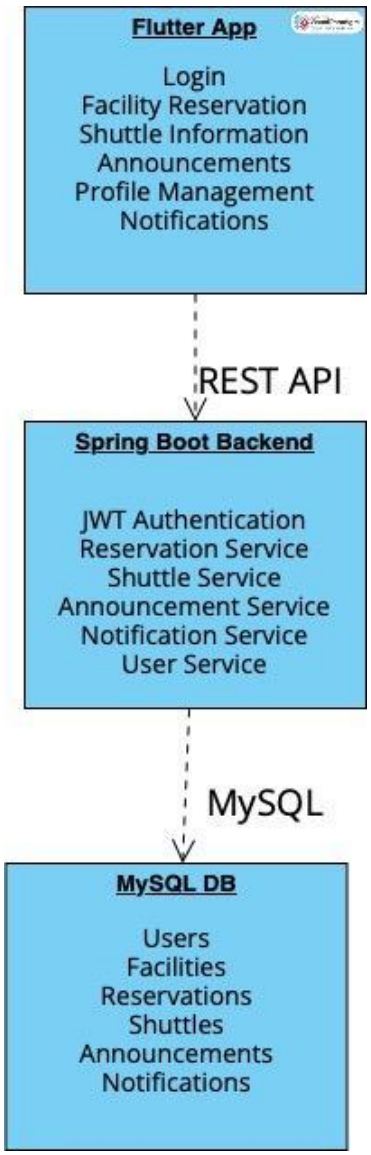


Figure 1. SMART-C System Architecture

3.2. Graphical User Interface

The graphical user interface of SMART-C is developed by using Flutter. The interface is designed to make accessing easier to services of campus. The application has a user-friendly design which allows all users to navigate between modules quickly and simply.

- Login Screen

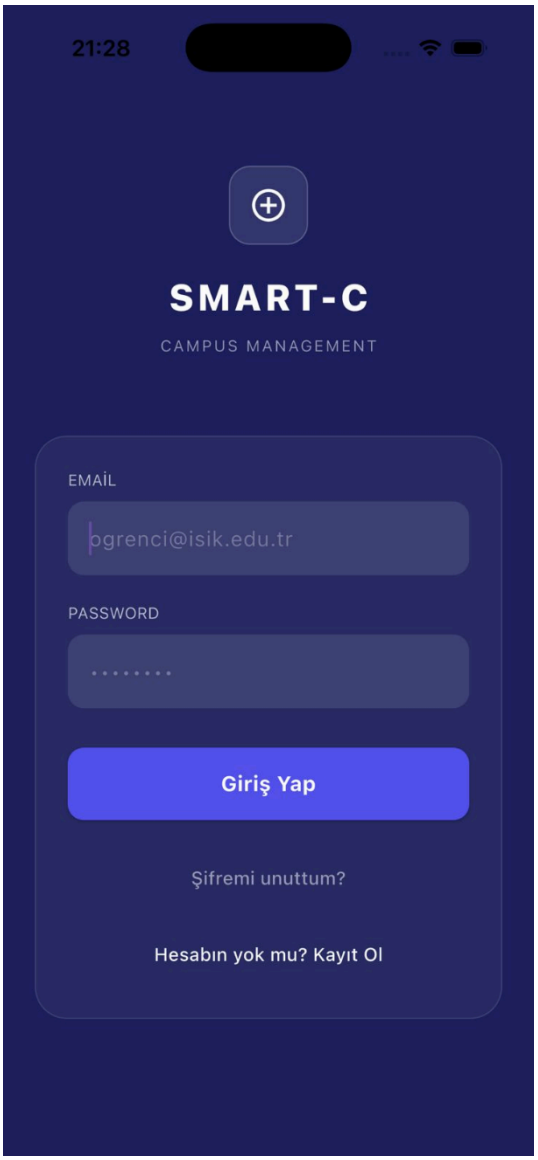


Figure 2.Login Screen

- Registration Screen

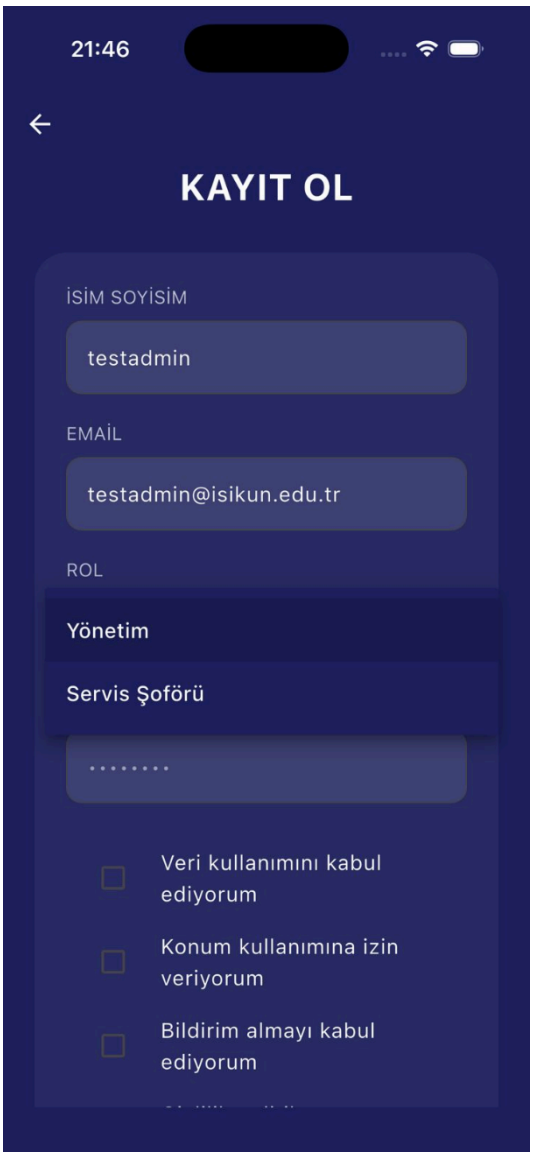


Figure 3.Registration Screen

- Forgot Password Screen: If the user who is already signed up forgets his password , he clicks the button and navigated to here. And the mail goes the users e-mail.

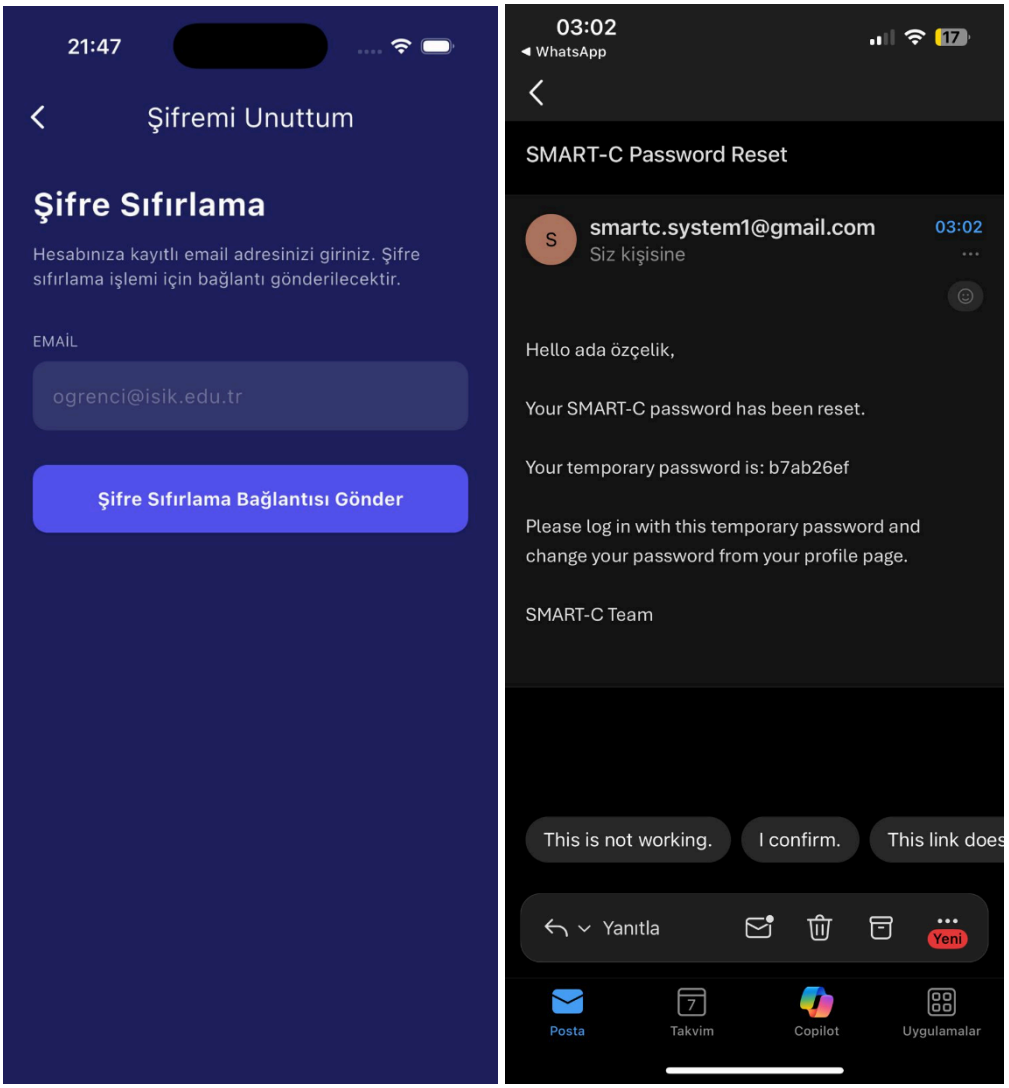


Figure 4.Password Resetting

- Dashboard: The main screen .



Figure 5.Dashboard

- Shuttle Screens : The shuttle screen has different interfaces for students, admins and shuttle admins. If the user is student or admin then he will see the right one , if the user is shuttle admin then he will see the left one only to make announcements.

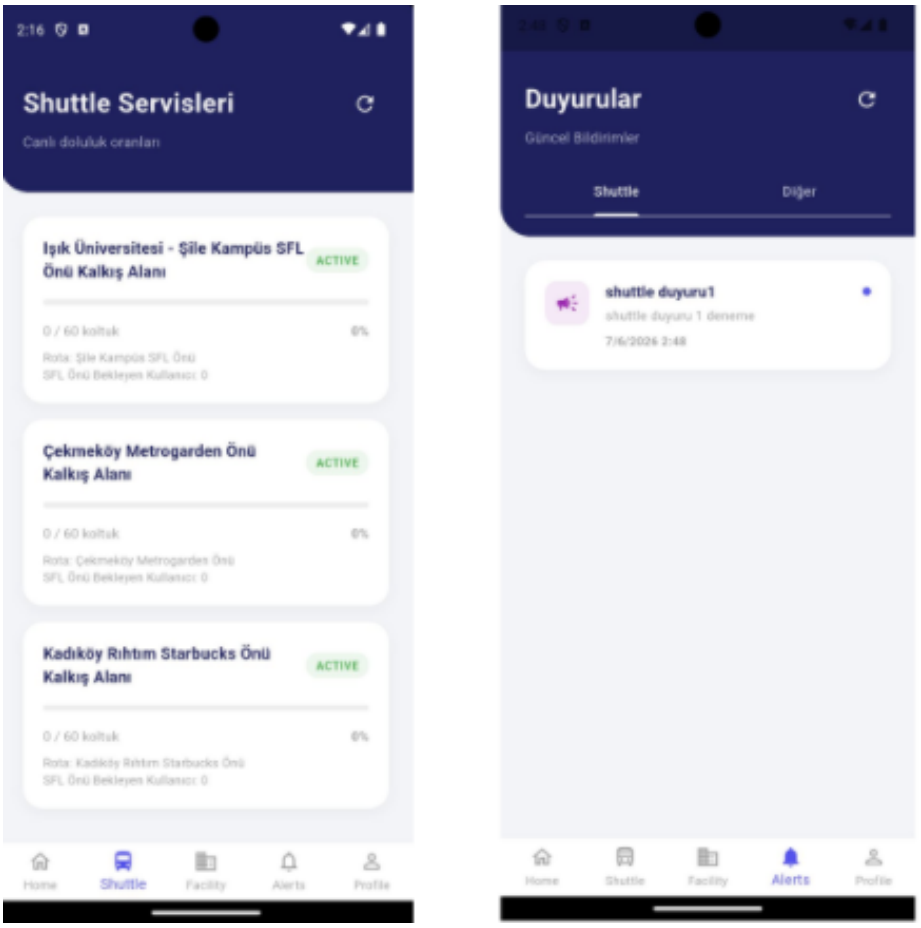


Figure 6. Shuttle Screens

- Facility Screen: For users which are other than shuttle admins , we have facility screen to reserve and see the availability .

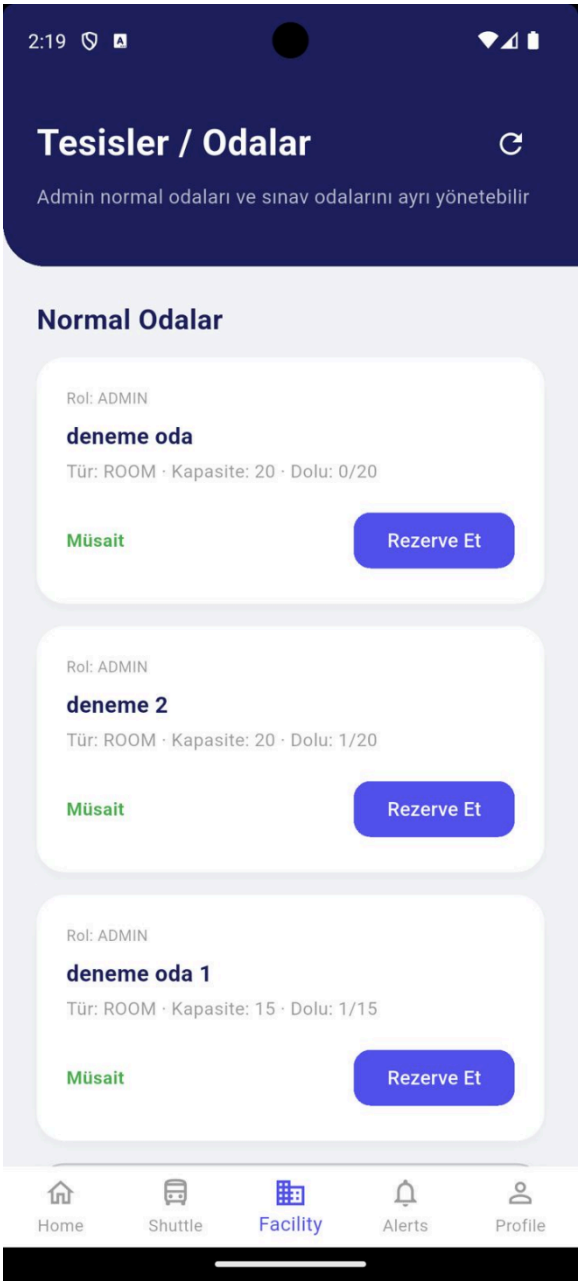


Figure 7.Facility Screens

- Adding or Deleting Facility Screen: For admins other than shuttle admins, the app has a screen for deleting or adding facility like classes to be reserved .



Figure 8.Adding Facility Screen

- Creating Facility Screen: Again for admins other than shuttle admins, to create a facility there is a pop-up screen with some information requests.

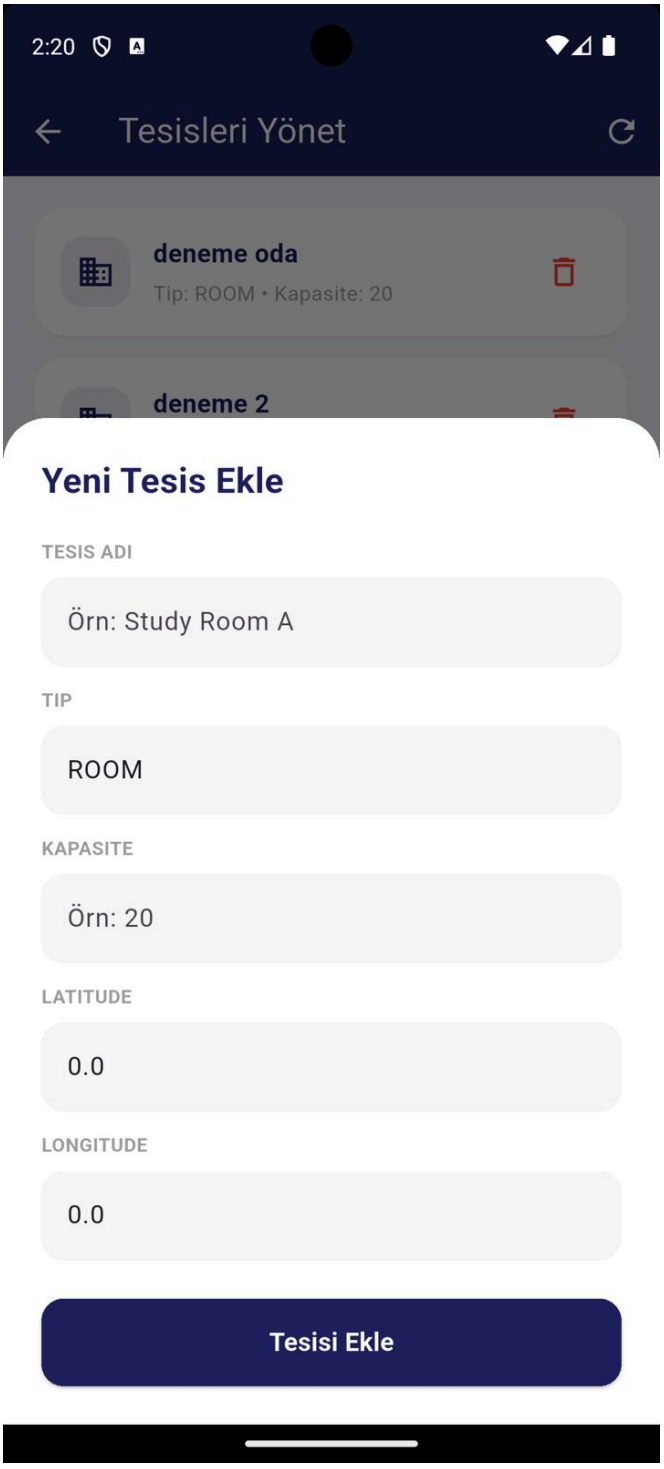


Figure 9.Creating Facility Screen

- Announcement Screen:The screen shows the current announcements.

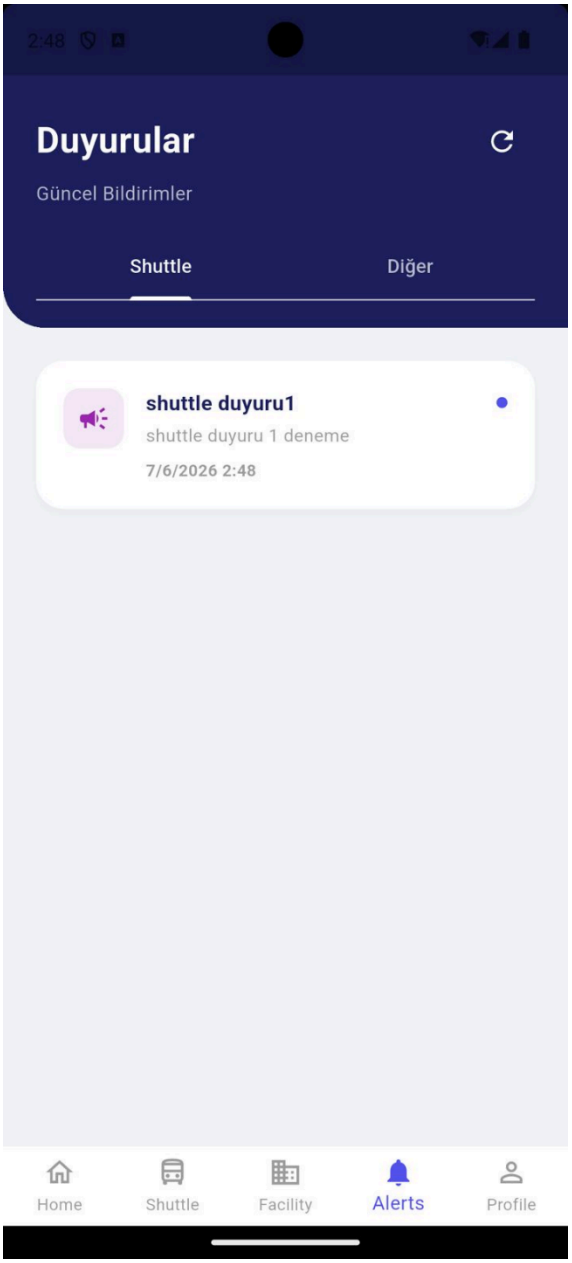


Figure 10.Announcement Screen

- Create An Announcements Screen: To create a new announcement only admins have a button on profile management screen . Not students.



Figure 11.Create Announcement Screen

- Settings Screen: The setting screens for admins and students . They have their roles as well.

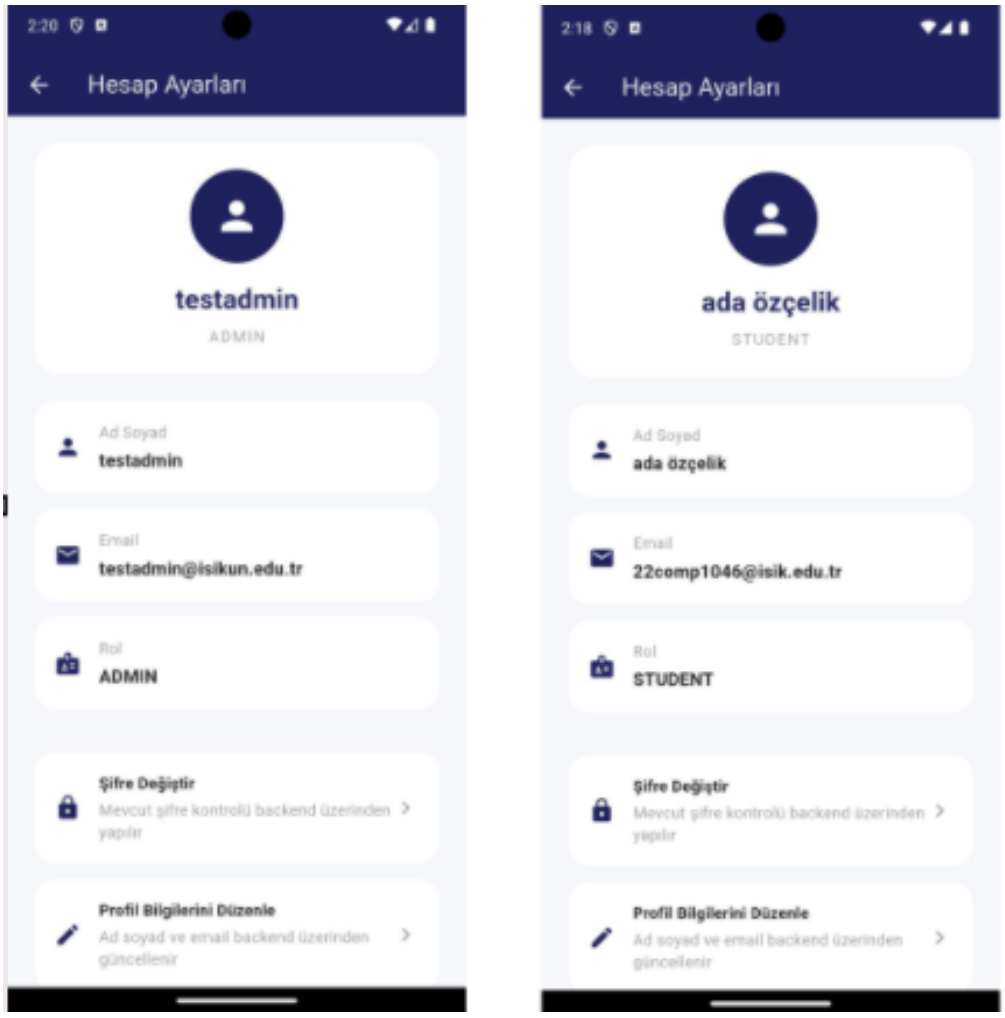


Figure 12.Setting Screen

- Privacy Screen: All of users have same privacy screen to access permisions.

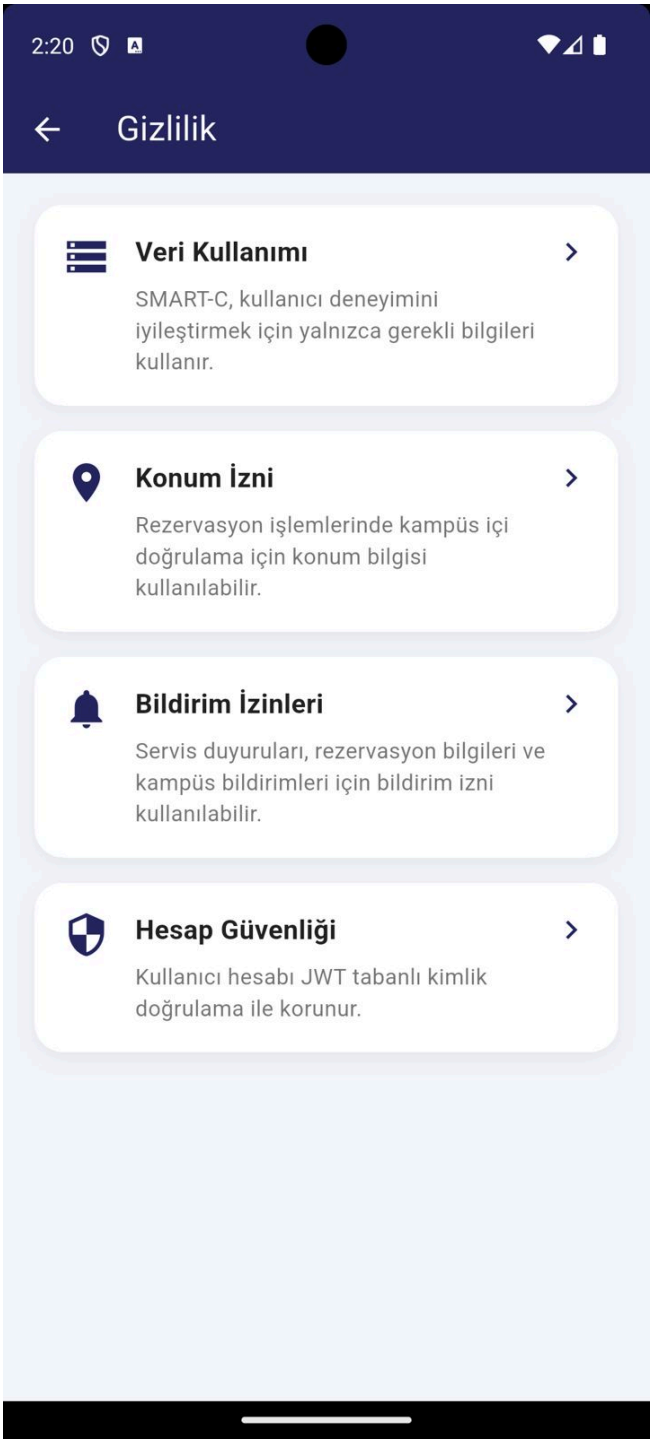


Figure 13.Privacy Screen

3.3. Business Logic

The business logic layer of SMART-C provides the primary operational policy of the system and manages the interaction among users, functionalities of the system, and the backend services. It involves user authentication, reservation management, shuttle management, announcements management, and user profile management. All functionalities are performed as per the authorized user roles that include Student, Admin, and Shuttle Admin.

Figure 14 depicts the primary use cases of SMART-C and their interaction with the user roles and functionalities of the system.

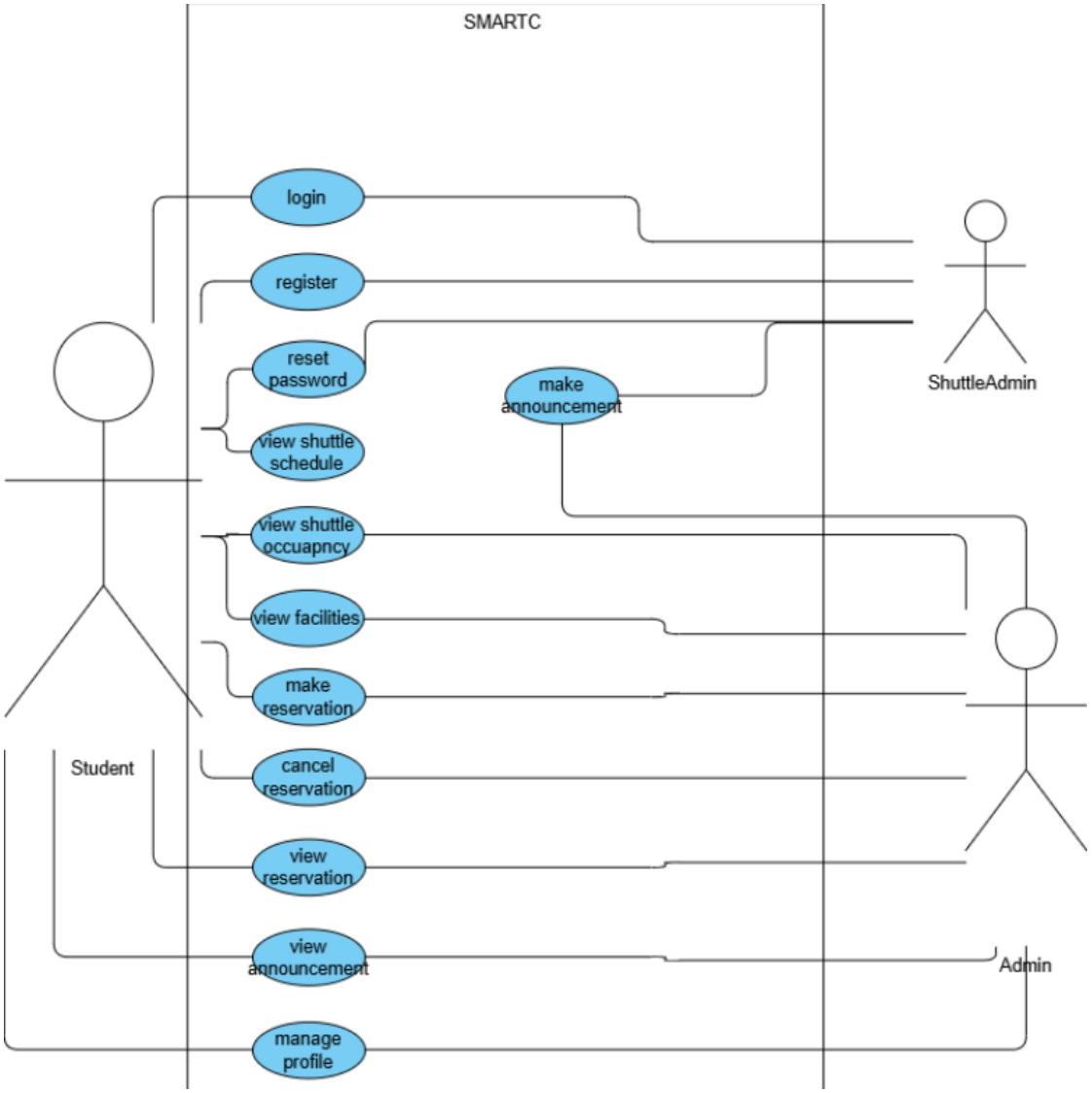


Figure 14.Use Case Diagram

3.3.1. Login Services

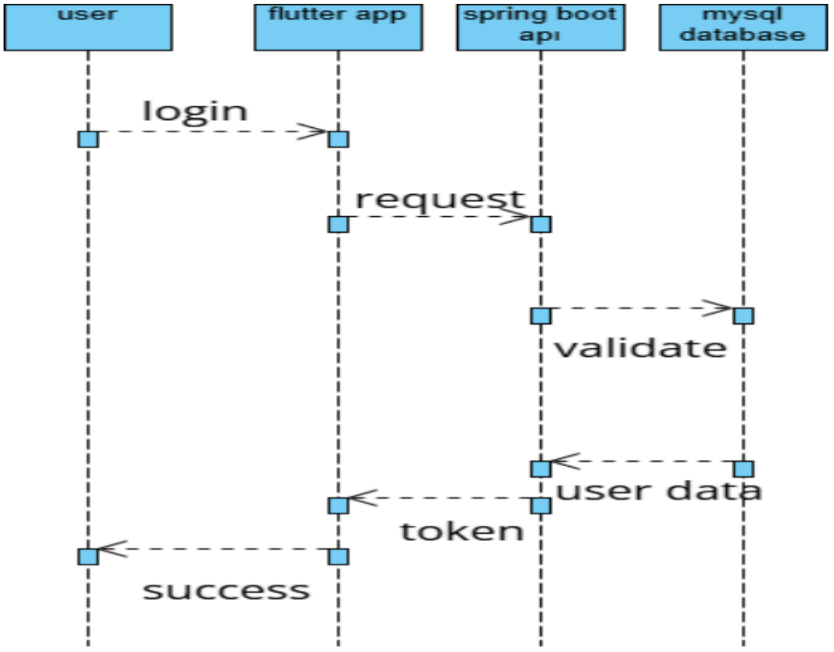


Figure 15.Login Authentication Process Sequence Diagram

The system has an efficient method of controlling access to the system resources by providing a secure way of logging into the system and accessing system functionalities based on the users' roles.

Users have the option of registering for new accounts in which they can log in using their credentials and access functionalities within the system. During the registration process, the system receives information from users concerning personal details, email, password, and role of each individual. The information provided is validated before creating the account.

Whenever users decide to access any resource in the system, their credentials are authenticated via the RESTful API of the application. If authentication is successful, the server generates and returns to the client a JSON web token (JWT).

After receiving the JWT from the server, the mobile application stores the token locally using the Shared Preference library. The token is sent to the server on every subsequent call made from the mobile application, thus verifying user identities and allowing the backend server to authorize access based on user roles.

The login service has a feature that helps recover passwords. Users may send password reset request whenever they are unable to recall their login passwords.

3.3.2. Shuttle Services

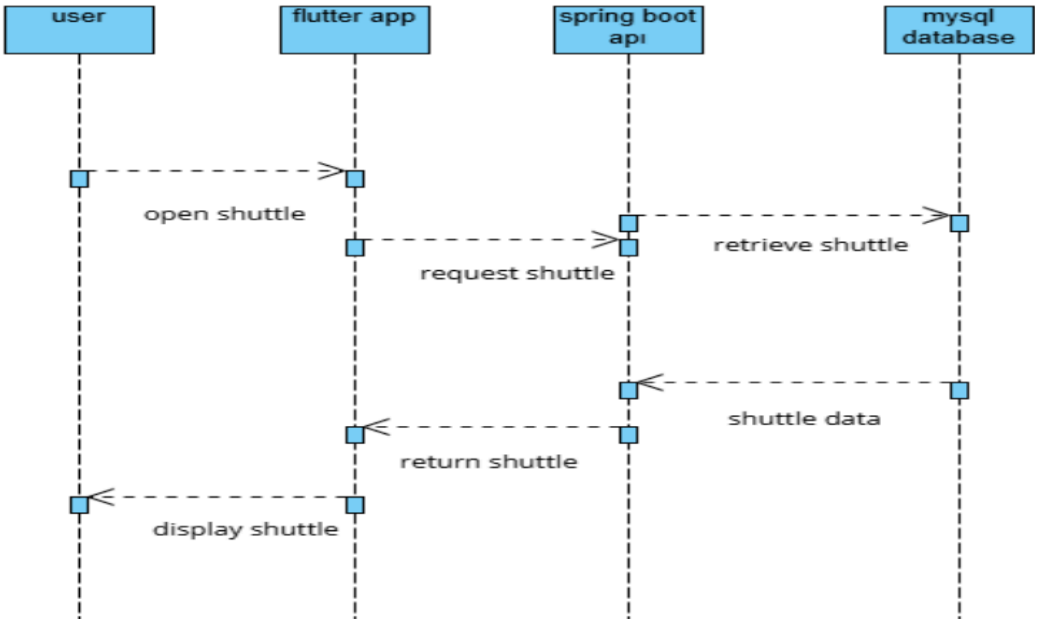


Figure 16.Shuttle Information Retrieval Sequence Diagram

The transportation module is responsible for storing the transport-related information in the SMART-C system. The information about shuttles, as well as other transport-related announcements, can be accessed by students via the mobile application.

Individuals who have the SHUTTLE_ADMIN permission can manage transportation announcements, such as delays and schedule changes, by using the proposed approach. All transportation announcements will be stored in the database and then distributed among users using the app.

The transportation module allows one to implement the location-based approach to estimating shuttle occupancy as presented in SMART-C. This can be accomplished with the use of location data from the mobile phones of authenticated users located close to shuttle departure locations.

The functionality of the transportation module includes all the operations performed via the RESTful API of the Spring Boot back-end.

3.3.3. Facility Services

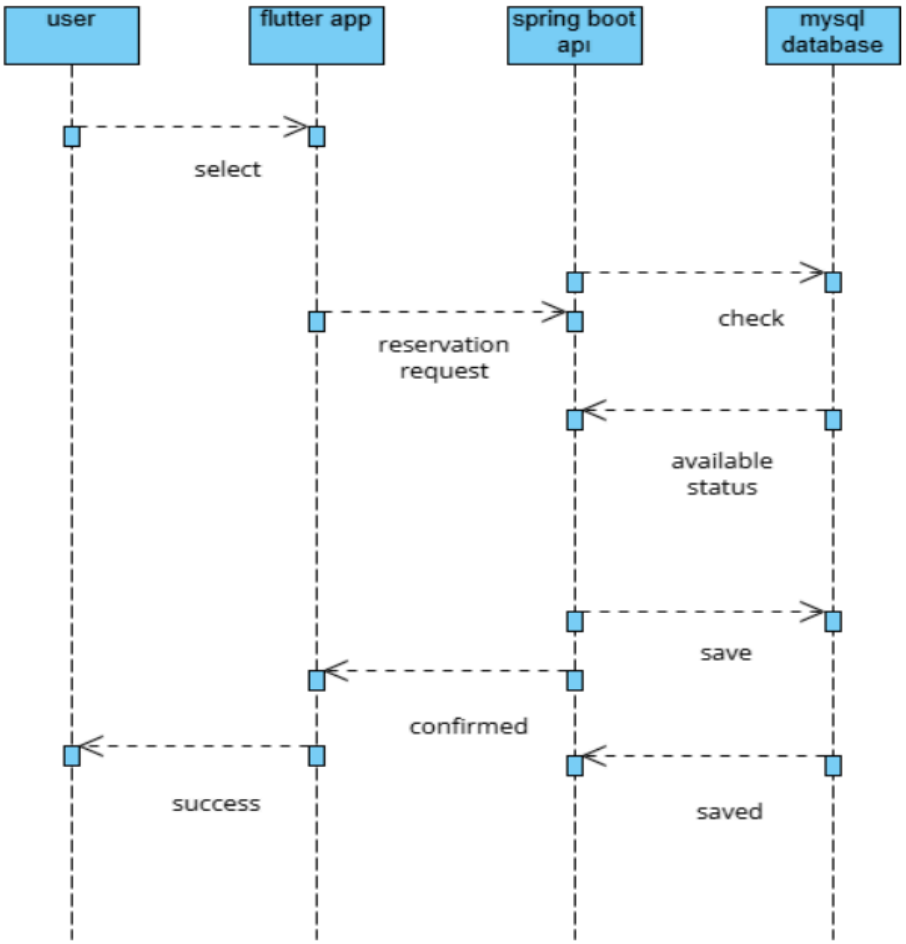


Figure 17.Facility Reservation Sequence Diagram

The facility service module allows users to make reservations for facilities on campus using the SMART-C mobile application. Users will be able to see the available facilities on campus and book them as per their requirements.

Prior to approval of the reservation request, there will be a validation process to check whether or not the chosen facility is available at that particular time interval. In case of a schedule clash, the user will not be allowed to make a reservation.

Administrators will have the ability to add/delete any facilities. There will be provision of exam reservations for authorized personnel to reserve classrooms for exams without any scheduling clashes.

Facility operations will all take place through the Spring Boot backend, which will further store them in the MySQL database.

3.4. Data Management

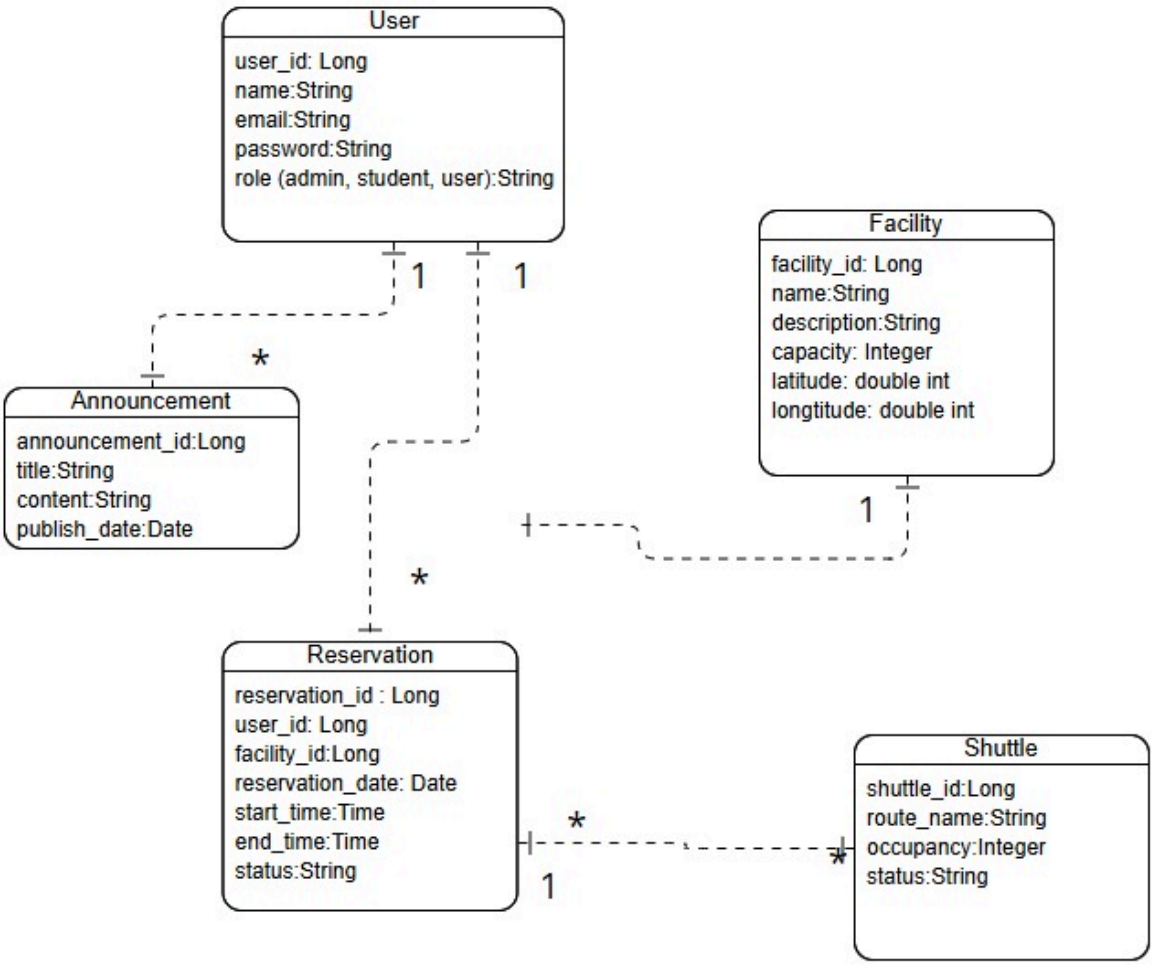


Figure 18.Entity–Relationship (ER) Diagram of the SMART-C Database

Figure 18 illustrates the Entity–Relationship (ER) diagram of the SMART-C database.

This database structure ensures data consistency and supports the main functionalities of the SMART-C system.

- **User Entity**: User information is held in the user entity. Each user has a unique ID, his or her name, email address, encrypted password, and role information. User entities are related to reservation and notification entities.
- **Facility Entity**: The Facility entity represents campus resources that can be reserved by users. Each facility contains information such as facility name, type, capacity, latitude, and longitude.

- **Reservation Entity**: The Reservation entity holds the reservation details that are made by users. It has one reservation detail per user and per facility. This entity holds information about reservation starting time, ending time, status, and reservation type.
- **Shuttle Entity**: The Shuttle entity stores information about transport. Each shuttle contains data on routes, capacity, occupancy, and availability status. The Shuttle entity enables students to view transport information using the mobile app.
- **Announcement Entity**: The Announcement entity manages campus announcements published by administrators. Each announcement contains a title, content, and creation timestamp.

4. IMPLEMENTATION

In this section, the implementation of SMART-C system will be elaborated. The SMART-C system has been developed according to client-server architecture using a mobile application based on Flutter, a Spring Boot backend service and MySQL relational database. Implementation objectives included providing a centralized platform for users to access shuttle information, make facility reservations, view announcements, receive notifications, and perform administrative tasks.

Specific focus was made on usability, scalability, maintainability and security issues when designing SMART-C system. A mobile application has been developed using Flutter framework, while business logic and data management functionality has been implemented using Spring Boot framework. All the system data is stored in MySQL database. RESTful API has been used for interaction between mobile application and backend. Security issues have been addressed using JWT (JSON Web Token) authentication and authorization.

Flutter framework has been chosen due to its capability of developing cross-platform applications which work both on Android and iOS platforms. Thus, it saved us time, reduced maintenance efforts and provided similar user experience for both iOS and Android.

The technologies and tools used in the implementation process are presented in Tablo 2.

Tablo 2.SMART-C Technology Stack

Component	Technology / Tool	Purpose
Mobile Application	Flutter	Cross-platform mobile application development for Android and iOS
Programming Language	Dart	Mobile application development
Backend Framework	Spring Boot	Development of RESTful web services
Programming Language	Java	Implementation of business logic
Database	MySQL	Storage of system data
ORM Framework	Spring Data JPA / Hibernate	Database mapping and data access operations
Security	Spring Security & JWT	Authentication and authorization
Location Services	Geolocator	Obtaining user location and validating reservation distance constraints
API Communication	RESTful APIs	Communication between mobile application and backend
Local Storage	Shared Preferences	Storing JWT tokens and user information on the device
Email Services	Spring Mail	Password reset and system email notifications
Development Environments	Android Studio, IntelliJ IDEA, Visual Studio Code	Mobile and backend development
Testing Environment	Android Emulator, iOS Simulator	Cross-platform application testing

4.1. Implementation Details

4.1.1. System Architecture and Development Environment

The implementation of SMART-C involved the use of a client-server architecture with multiple layers. In particular, a Flutter application, Spring Boot backend, and MySQL database were utilized. The choice of the architecture allowed for scalability and maintainability as well as ensured a good separation of roles in the system.

The presentation layer of the system consists of a mobile application. It is written in Flutter using Dart programming language and hence allows running it on both Android and iOS platforms using one source code. Its main role is to display various data related to shuttle transportation (information about shuttles, facilities, announcements, and notifications) and receive user inputs.

The backend was built on the basis of the Spring Boot framework and uses layered architecture (controllers, services, repositories, and entities). Controllers retrieve requests from mobile applications via RESTful API and forward them to the service layer which

implements the business logic of the system (authentication, reservation check, shuttle management, facility management, and notifications).

The data of all system entities have been saved in a MySQL relational database. In this database, there are tables corresponding to users, facilities, reservations, shuttles, announcements, notifications, etc. The Hibernate ORM automatically maps the entities in the Java domain model to database tables, decreasing manual SQL code needed for their work.

In order to ensure security, Spring Security along with JWT authentication was utilized. When the user logs in successfully, the backend issues a JWT and sends it to the mobile app. This token is stored locally in Shared Preferences and added to further API requests. Thus, only authorized users may access the protected services.

Geolocation services were realized utilizing the Geolocator plugin. The mobile app receives the current user's location and sends these coordinates to the backend when needed.

Figure 19 illustrates the overall architecture of the SMART-C system and the interaction between its major components.

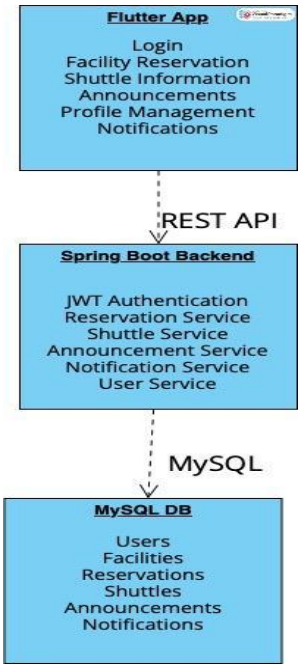


Figure 19.SMART-C System Architecture

4.1.2 Authentication and User Management Implementation

Authentication and user management features have been developed in order to guarantee secured access to SMART-C platform.

The system provides the following features: user registration, user logging-in, password reset, user roles management, and session handling.

Registration includes filling out personal details, specifying the email, password, and role for which the registration is being done. Backend checks the input data, and then saves them in the MySQL database. User passwords are hashed in order to prevent any access to them from outside of the backend.

User login functionality has been realized with help of Spring Security framework and JWT (JSON Web Token) authentication. After successful login, backend creates a JSON web token which is then sent to a mobile app. That token is saved using Shared Preferences on a client side, and sent to API in the Authorization header in order to authorize further requests.

The system implements role-based access control to restrict specific operations according to user roles. Three primary roles are supported within SMART-C: STUDENT, ADMIN, and SHUTTLE_ADMIN. Students can create and manage reservations, view announcements, and access shuttle information. Administrators can manage facilities, announcements, exam reservations, and user-related operations. Shuttle administrators are responsible for shuttle management and transportation-related announcements.

A password recovery mechanism was also implemented to improve usability. Users who forget their passwords can request a password reset through the mobile application. The backend generates a password reset process and sends the necessary information to the user via email using Spring Mail services.

By combining JWT authentication, encrypted password storage, email-based password recovery, and role-based authorization, the SMART-C system provides a secure and reliable user management infrastructure.

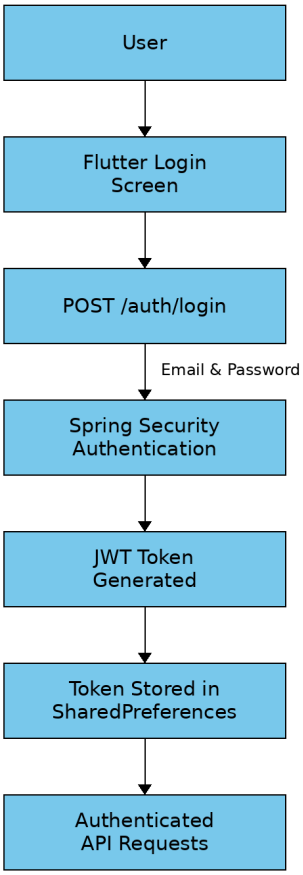


Figure 20. JWT Authentication Flow

4.1.3 Reservation Management Implementation

Reservation management is one of the crucial modules of the SMART-C system. The module allows users to make reservations for different facilities on campus in such a way that the efficiency of resources use is maximized and conflicts between schedules are avoided.

In the mobile application, users can find out which facilities are free and proceed to make reservations of them. When the reservation is made, it is sent to the backend server via RESTful APIs. After sending, it goes through several stages of validation.

Among those stages of validating reservation requests, the most essential is conflict checking. At the stage, the system makes sure there are no conflicting schedules existing in relation to the chosen facility within the proposed time frame. If there are any conflicts, the reservation will not be accepted, thus preventing the creation of duplicate reservation records.

Moreover, it checks if the chosen facility is present in the database and available for booking.

For location-based functionalities, the Geolocator package was implemented on the mobile application platform. When required, the location of the user is retrieved and forwarded to the server. The distance between the coordinates of the chosen facility and the user is then calculated. This feature plays a key role in reservation validation processes and helps prevent the misuse of facilities in the campus.

Apart from ordinary reservations, the system has implemented the exam reservation feature for authorized administrators. Using the system, administrators are able to book particular classes for conducting exams without causing any overlapping of exam bookings for a single class. Thus, booking processes of classes become systematic.

Reservation records in the system are stored in the MySQL database together with the relevant facility and user records. Users can check their reserved facilities using the mobile application. On the other hand, the administrator can monitor reservation activities.

This way, booking validation, conflict prevention, location checking, and administration contribute significantly to efficient facility usage.

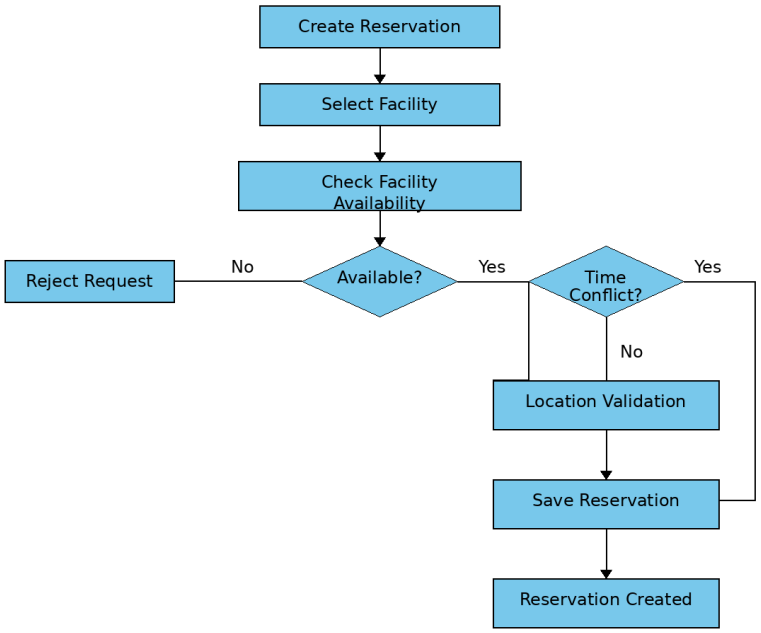


Figure 21.Reservation Validation Flow

4.1.4. Shuttle Management Implementation

The development of the shuttle management module aimed at providing transportation information to students as well as enhancing the management of shuttle services. Users have access to shuttle information, while shuttle administrators will be responsible for managing transportation services in the system.

Information regarding shuttles is kept in the MySQL database, which is controlled via the Spring Boot back end. The back end supplies RESTful API endpoints that enable a mobile application to obtain and present information related to shuttles. The use of such an approach ensures that the user gets access to the latest information on transportation services offered.

A number of roles are created to serve different purposes within the shuttle management module. The first one refers to students who need to see shuttle information and announcements about transportation in the mobile application. Administrators of shuttles have other responsibilities that include managing transportation announcements.

The development of the module was done to implement the transport concept introduced by SMART-C. The integration of information about shuttle buses and the user's location using the Geolocator library allows for making decisions that concern transport and providing additional information about transportation facilities on the campus.

The mobile application provides information on transportation facilities using an intuitive graphical interface. All shuttle-related announcements made by authorized administrators are displayed in the application itself, allowing for successful communication between transport managers and students.

All information and announcements related to shuttle bus transportation are managed by the central back-end system.

With the help of information about shuttle transport, the role-based management of the information, location-based approach, and the ability to make announcements, the shuttle management module helps streamline the process of transportation on campus.

4.1.5. Announcement and Notification Implementation

The announcement/notifications module was developed with the purpose of facilitating better communication between the campus administration and its users. Through this module, it becomes possible to spread important information on transportation, facility, reservations and other campus-related activities using one unified platform.

Admins authorized to use the system will have access to management modules where they will be able to create and edit their announcements. After posting the announcement, it will become visible for all users of the app. The usage of only one channel for spreading important information is another advantage of the proposed solution.

Moreover, the system allows for organizing announcements according to roles. Thus, general campus announcements can be created by admins while shuttle administrators will post information concerning transportation issues.

Notifications are sent to alert users of any system event and activity. They include reservation confirmation, reservation cancellation, transportation, and administrative announcements. In this regard, the system makes sure that users stay informed about the latest developments in their activities at the campus.

Announcements and notifications will be recorded on the MySQL database and will be managed via the Spring Boot backend. Communication for transferring announcement and notification data is done through the use of RESTful APIs. This ensures that the data being transmitted is consistent.

The mobile app will have specific pages where users can see announcements and notification related information. The user interface makes it easy to see any information in relation to announcements and notifications without using another communication tool.

Announcement management and notification services have been combined in one package in the SMART-C solution.

4.1.6. Mobile Application Implementation

The SMART-C mobile application was written using the Flutter and Dart programming languages. The Flutter library was chosen because it supports cross-platform development. In other words, only one codebase is required to build applications for Android and iOS devices. This strategy saves time during the development process and maintenance stage.

The application contains several screens which provide access to all basic functionalities of the system such as user authorization, facility booking, shuttles info, announcements, notifications, account management, and administrative tools. All transitions between screens are supported by the application router, provided by Flutter.

APIs were used to implement communication between the frontend and backend parts of the system. RESTful architecture was adopted, while the communication with back-end resources was based on a dedicated API service which managed all requests and responses via HTTP protocol.

In order to keep user sessions, JWT tokens and user data were stored on the client side in a shared preferences file. After authentication, the token is cached and sent automatically with every request made by the API.

Geolocator plugin was incorporated in order to implement functions related to locations. It allows obtaining the user's current location whenever necessary, and transmitting this information to the backend server in order to validate reservations and perform transportation-related actions.

Usability and user interface aspects received special consideration at the stage of application development. The application was intended to provide convenient access to campus facilities without requiring the user to undertake too many actions, thus optimizing interface design by implementing consistent layout and navigation components as well as role-based interfaces.

The mobile application acts as an interface between users and SMART-C system, facilitating access to campus facilities via mobile devices.

4.1.7. Database Implementation

MySQL is used in the implementation of the SMART-C system as the database management system. Database operations involve storage and management of data that include user data, facility information, reservation, shuttle information, announcements, and notifications. Relational database is chosen because it ensures data consistency, data integrity, and efficient data retrieval.

Spring Data JPA and Hibernate frameworks were used in implementing database operations. This framework offers an Object-Relational Mapping (ORM) which maps the entities in the application into the database table. Therefore, all database operations are executed without involving writing of SQL code since it is handled by the framework.

User information is stored in the database in relation to user entities. The information stored includes personal information about the user, authentication data, and roles.

The Facility entity consists of the details of reservable items in the campus like classroom, study rooms, laboratories, and other facilities. Each facility record holds information on facility name, facility type, facility capacity, and facility coordinates.

The Reservation entity consists of reservation records created by users. Each reservation record is tied with a unique user ID and facility ID, thus helping the application in tracking the reservation activities.

The Shuttle entity maintains information about transportation. The Shuttle records hold information on shuttle name, shuttle occupancy, and other necessary details to manage the shuttle management system.

The Announcement entity consists of announcements created by administrators and shuttle administrators. Each announcement contains a record title, content, publish date, and other information required for information management.

The Notification class is responsible for holding system generated notifications and specific user messages. This architecture helps in sending important notifications and information regarding reservations to the users.

Relationship between entities was achieved by making use of JPA annotations. These annotations include relationships like One-to-Many, Many-to-One, and JoinColumn which make sure that data is efficiently managed.

MySQL along with Spring Data JPA and Hibernate makes the database layer both robust and scalable while fulfilling all the basic requirements of the SMART-C system.

5. TESTS AND EXPERIMENTS

5.1. Acceptance Tests (Functional)

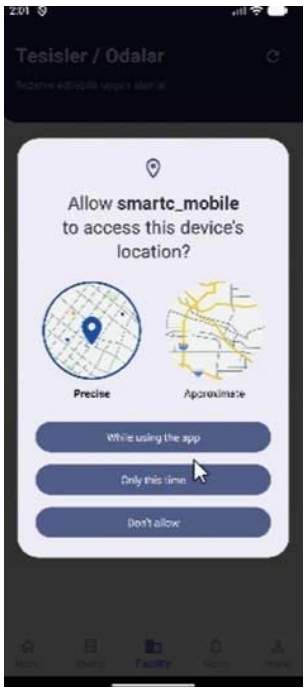
Acceptance testing was conducted to be sure that the SMART-C system satfies functional requirements defined on RAD.

Tablo 3.TC

Test ID	Test Scenario	Expected Result	Actual Result	Status
TC-01	User Login	User successfully logs into the system	User successfully logged in	PASS
TC-02	Invalid Login Attempt	Login request rejected	Login request rejected	PASS
TC-03	View Shuttle Information	Shuttle information displayed correctly	Shuttle information displayed correctly	PASS
TC-04	Create Reservation	Reservation created successfully	Reservation created successfully	PASS
TC-05	View Reservation	User reservations displayed correctly	User reservations displayed correctly	PASS
TC-06	Cancel Reservation	User cancels reservation successfully	User cancels reservation successfully	PASS
TC-07	Login Restriction Test	Reservation request denied outside authorized area	Reservation request denied	PASS
TC-08	Valid Location Reservation	Reservation request accepted inside authorized area	Reservation request accepted	PASS

TC-09	Student Announcement Creation Attempt	Student cannot create announcements	Operation denied	PASS
TC-10	Admin Access	Admin can access announcement management	Access granted	PASS
TC-11	Reservation Conflict Prevention	Duplicate reservation prevented	Reservation rejected	PASS
TC12	Create Announcement	Announcement successfully created	Announcement successfully created	PASS
TC-13	View Announcement	Published announcement visible to users	Announcements displayed	PASS
TC-14	Logout	User successfully logs out	User successfully logged out	PASS





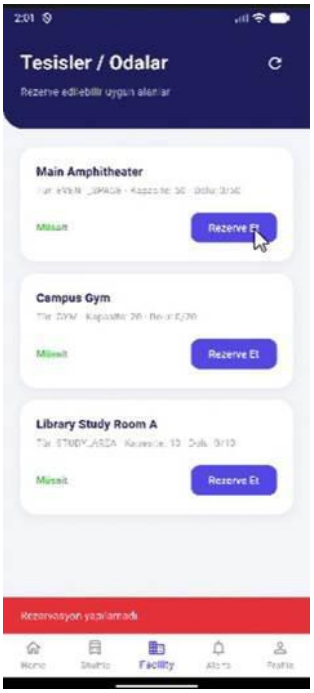


Figure 22Login denied because of invalid info

Figure 23.App wants to know location

Figure 24.App doesnt let to reservate user who is not in valid area

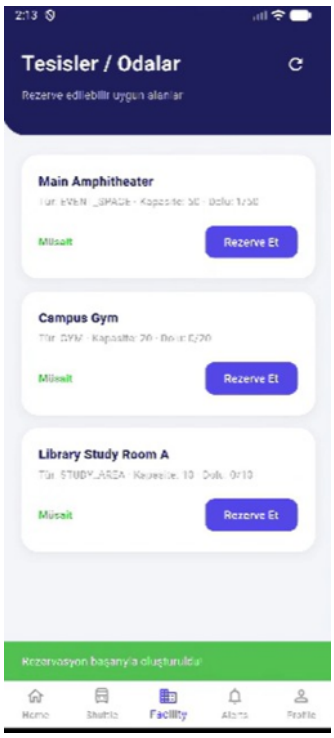


Figure 25. App lets user reserve who is in the valid area.

5.2. Non-Functional Tests

The SMART-C was also evaluated according to non-functional requirement.

- **Usability**: The SMART-C application provides a simple mobile interface. Users can access shuttle information , make reservations and view announcements with small number of interactions. The developed interfaces minimize navigation complexity.
- **Reliability**: The reservation management mechanism prevents double-booking of facilities. Reservation information is stored persistently in database. The system maintains consistency between application , backend, database.
- **Performance**: System operations like login , viewing shuttles , managing reservation are completed within response times. Changings made by admins become visible to users.
- **Security**: Authentication mechanisms makes sure that only registered users can access system. Role-based authorization prevents unauthorized users from accessing administrative functions. The local validation mechanism provides a security layer for operations.
- **Supportability**: Administrative interfaces allow authorized users to manage announcements and facility information without modifications to source code.

5.3. Experiments (Optional)

Several practical experiments are finalized to validate critical system behaviors.

The first experiment evaluated the location-based reservation mechanism. Reservation requests submitted outside the area authorized area were rejected while requests submitted from valid locations were accepted successfully.

The second experiment evaluated role-based authorization. Student accounts were unable to access announcement management features, where administrator accounts successfully performed creation operations.

The third experiment evaluated reservation conflict prevention. When a user attempted to reserve a facility which had already been reserved for selected time slot , the system rejected request and prevented reservations

6. CONCLUSIONS AND FUTURE WORK

6.1 Conclusion

In this project , SMART-C (Smart Management and Reservation Tool for Campus) is developed to boost or improve campus transportation and facility reservation process. The system is served to users with a mobile application for viewing , managing,reservation so many things about campus currently.

The application was implemented using Flutter , Spring Boot and MySQL.results of testing showed that the main functionalities, including authentication, reservation management, announcement management , location-based validation.

6.2 Future Works

Future versions of SMART-C may include real-time shuttle tracking, push notifications, QR -code -based reservation verification and advanced reporting features administrators.

Additional campus services like parking area , event management , visitor access control may also be system to further improve campus operations.

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